APPLICATION OF STATISTICS IN TQM

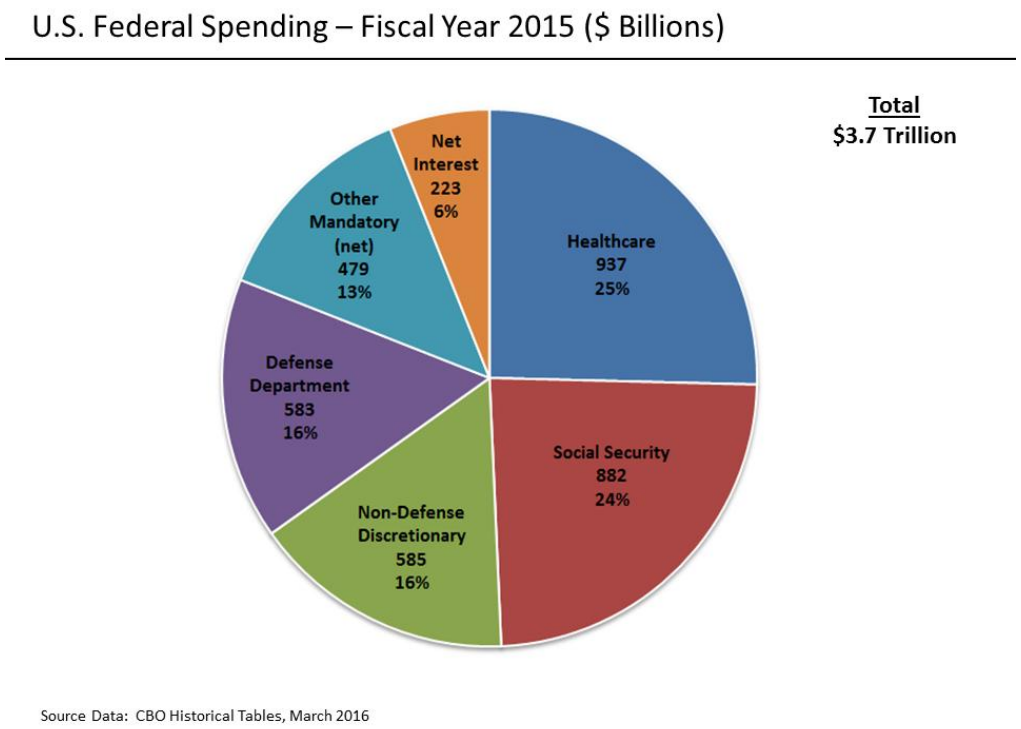
Twelfth Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati

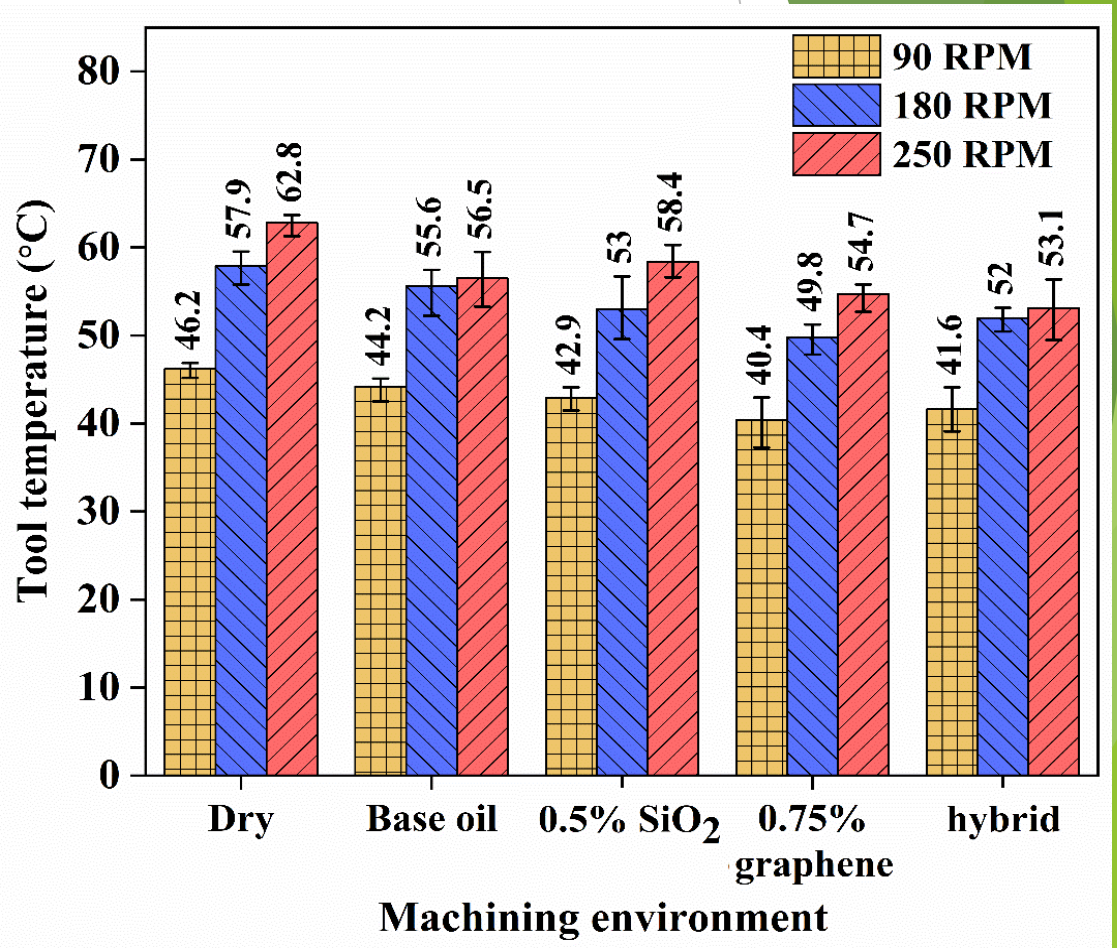


Data=measurement or observation (numbers)

Presentation of data

Frequency diagram, bar chart, pie diagram, frequency graph, Line diagram, frequency curve





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Types of statistics:

Descriptive, Inferential

Examples of Descriptive statistics:

- Average rainfall in Shillong last year
- Number of new phone connections in 2003

What are Statistics?

- A set of methods (mathematical procedures) for organizing, summarizing, presenting and interpreting information.
- Help bring order out of chaos
- Provide standardized techniques for communicating and interpreting information
- Vocabulary and symbol for communicating about data.
- A theory about decision making
- Tools that allow us to make “judgment under uncertainty.

Population versus samples

Population: Set of all individuals of interest in a study.

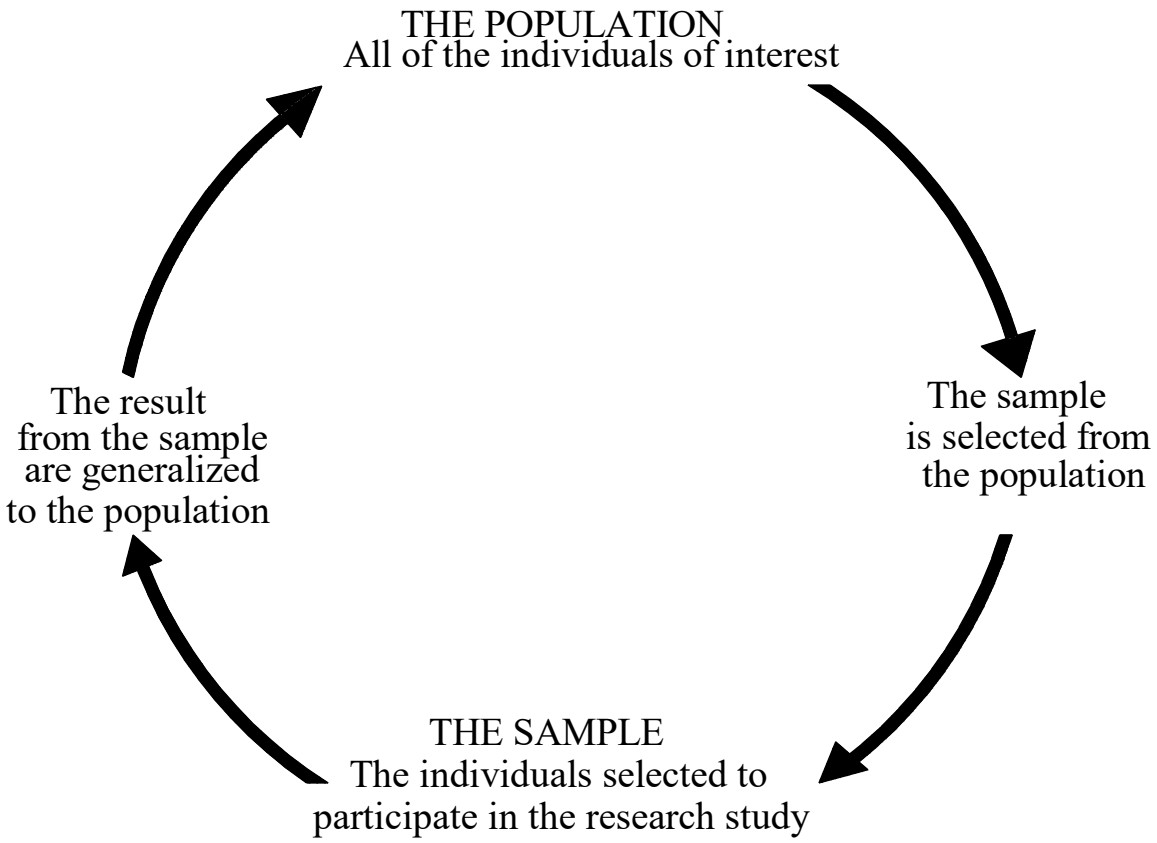
Sample: Set of all individuals selected from a population

- Intended to represent the population
- Should be identified in terms of the population from which it was selected

Why not just collect data from the whole population?

- Sometimes impractical
- Often impossible
- Usually unnecessary

Relationship between population and sample



Parameter versus statistics

Parameter = A value (usually numerical) that describes a population

- May be derived from a single measurement or a set of measurement from the population

Statistic- A value (usually numerical) that describes a sample

- May be derived from a single measurement or a set of measurements from the sample

Each population parameter has a corresponding sample statistics.

Inferential statistics

Tools for drawing inferences (generalization) about a population from data collected from a sample.

Examples:

Opinion poll of 1000 Indians for knowing: Who will be the best Prime Minister?

Representative samples

- A goal of inferential statistics is to generalize from samples to populations.
- Generalization will depend on how well the sample represents the population.

Representative sample = a sample whose characteristics are similar to the population

Random sampling (random selection)= Process for obtaining a sample from population that requires that every individual in the population have the same chance of being selected for the sample.

SAMPLING DISTRIBUTION

Consider a finite population of 4000 students of a certain college. Of this 200 students are selected randomly and their mean weight is calculated. It was computed as x = 61 kg. If we draw 20 different samples of 200 students, instead of just one as we did previously, we can't expect the same mean weight of these 20 samples chosen randomly. It is due to sampling variability i.e. each sample drawn randomly from the same population differs from each other in their computed '**statistic**'.

The means computed is listed as

59,62,64,59,60,60,58,57,59,63,60,61,64,66,62,61,58,65, 62,62

Sampling Distribution

These 20 values form what is known as a part of the sampling distribution of a **statistic "mean" weight**. Therefore, the sampling distribution is the statistic calculated for a large number of random samples drawn from the same population.

Sampling Distribution

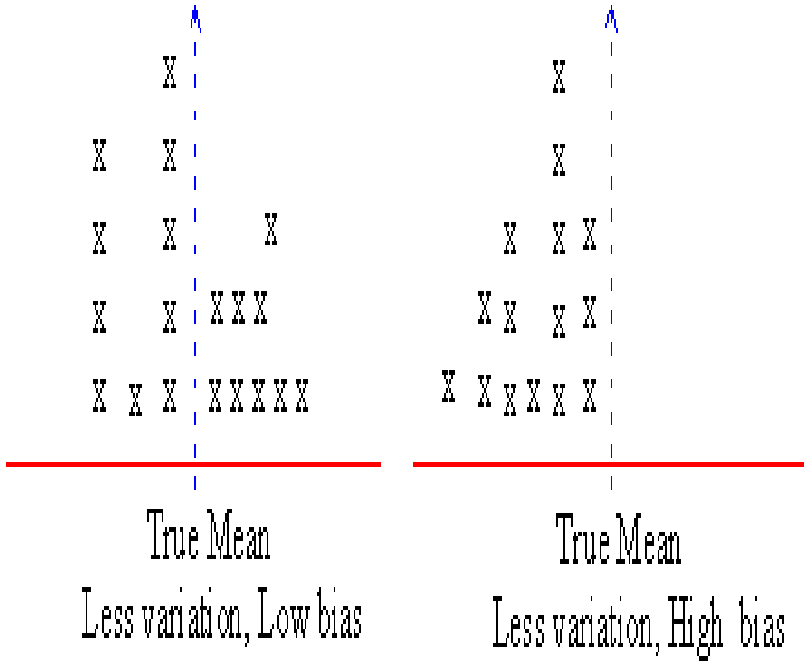
- Alternatively, consider all possible samples of size '*n*' which can be drawn randomly from the same population, and for each one, compute the aimed statistic. In this manner we obtain the distribution of the statistic, which is the sampling distribution.
- Now we compute mean of this sampling distribution which is 61.16 kg

Sampling Distribution

Now recall the list of means computed for 20 random samples. We observe that the 20 sample means differed form the true population mean m = 61.16 kg. The means 62, 64, 63, 60, 64, 64, 62, 65, 67, 62 are clearly above the true mean while 59, 59, 60, 60, 58, 57, 59, 60, 61, 61, 58 are below the true mean. These deviations or differences from true value is due to the variability of samples. This is known as 'Random error' in statistical estimation. Note that the mean of 20 random sample means is slightly greater (64 kg) than the true population mean. If we take 20 more samples of 200 students, the mean of this new sampling distribution might be lower than the true population mean.

Sampling Distribution

- The population mean (μ) i.e. mean weight of 4000 students is 61.16 kg (In case of infinite population, this is not known). Therefore, 61 kg is not a bad estimate of the population mean. But 61.15 kg is a still better estimation.
- The more the sample means (statistics) included in the sampling distribution, the more correct the chances of the mean of the sampling distribution becoming an estimate of the population mean (parameter).



- Other than random error, error arises due to the tendency of consistent under estimating or over estimating a true value. This is known as systematic error or bias. Suppose, in the example discussed above, an investigator is investigating only those students who are healthy and fit. He would most likely overestimate the population mean. Generally healthy bodies have reasonable weights. Clearly the investigator's choice is one of bias.
- This is an example from one of the bias that occurs in estimation.

Sampling Error

There is generally some difference between a statistic i.e. the value obtained from a sample and its corresponding parameter of the population. In example discussed, the mean of the weights of 200 students taken (first sample) is slightly different from the mean of the weights of 4000 students i.e. the population. This difference is called **sampling error** (whose types and situations are discussed). This error can be minimized by increasing the size of the sample and by taking sample of extremely random in nature. If the size of sample is large the sampling error is small. The accuracy of the sample increases as the square root of the size 'n' of the sample i.e. according to $\sqrt{n}$. Hence we take samples of sufficiently large in size.

Central Limit Theorem

What we have discussed with an example is stated as a theorem called **Central limit theorem**. It states that 'when the size of the sample increases and becomes sufficiently large the distribution of the sample means tends to Normal Distribution'.

Now here we must take one thing in our notice that the theorem is equally valid for both normal and strongly non-normal. We restate the central limit theorem as: "Even if the population distribution is strongly non-normal, its sampling distribution of means tends to normal distribution for large sample size".

- We have seen that the sample mean is distributed normally with mean *m*, the mean of the population, which is an unbiased estimator. It can be shown that the standard deviation of the sample means, called **Standard error (S.E.)** is equal to

$$\frac{\sigma}{\sqrt{n}}$$

Hypothesis testing

Many a time, we strongly believe some results to be true. but after taking a sample, we notice that one sample data does not wholly support the result. The difference is due to (i) the original belief being wrong and (ii) the sample being slightly one sided.

Testing of Hypothesis

Tests are therefore, needed to distinguish between the two possibilities. These tests tells about the likely possibilities and reveal whether or not the difference can be due to chance elements. If the difference is not due to chance elements it is significant and therefore, these tests are called tests of significance. The whole procedure is known is Testing of Hypothesis.

Hypothesis testing

- A hypothesis is a statement supposed to be true till it is proved false. It may be based on previous experience or may be derived theoretically.
- First a statistician or the investigator forms a research hypothesis that an exception is to be tested. Then he derives a statement which is opposite the research hypothesis (denoting as H_1).
- The approach here is to set up an assumption that there is no contradiction between the **believed result** and the **sample result** and that the difference therefore can be ascribed solely to chance. Such a hypothesis is called a **null hypothesis (H_o)**.
- It is the null hypothesis that is actually tested, not the research hypothesis. The object of the test is to see whether the null hypothesis should be rejected or accepted.

Hypothesis testing

- If the null hypothesis is rejected, it is taken as evidence in favor of the research hypothesis which is called as the **alternative hypothesis (denoted by H_α)**. In usual practice we do not say that the research hypothesis has been "proved" only that it has been supported.
- For example, if it is assumed that the mean of the weights of the population of a college is 50 kg, then the null hypothesis will be the mean of the population is 50 kg. i.e. H_o: m = 50 kg (Null hypothesis). In terms of alternative hypothesis (i) H_α : $m \neq 50$ kg (ii) H_α : $m > 50$ kg (iii) H_α : $m < 50$ kg

Setting up levels of significance

- Once the null hypothesis is set up, the next job is to set the limits within which we expect (the null hypothesis) *m* lies.
- The idea behind it is to ensure that the difference between the sample value and the hypothesis should arise due to sampling fluctuations alone.
- If this difference does not exceeds this limit then the sample supports the null hypothesis and the sample is accepted.
- If it exceeds this limit the sample does not support the hypothesis and it is rejected.

Hypothesis testing

- Now fixing the limits totally depends upon the accuracy desired. Generally the limits are fixed such that the probability that the difference will exceed the limits is 0.05 or 0.01. These levels are known as the 'levels of significance' and are expressed as 5% or 1% levels of significance. Rejection of null hypothesis does not mean that the hypothesis is disproved.
- It simply means that the sample values does not support the hypothesis. Also, acceptance does not mean that the hypothesis is proved. It means simply it is being supported.

Confidence limits

- The limits (or range) within which the hypothesis should lie with specified probabilities are called the confidence limits or limits. It is customary to take these limits at 5% or 1% levels of significance.
- If sample values lies between the confidence limits, the hypothesis is accepted; if it does not, the hypothesis is rejected at the specified level of significance.
- Confidence level is the complement of the level of significance. 5% level of significance =95% confidence level

An example

The following table gives the values of protein from the milk of a certain breed of cow and buffalo. It is to be examined whether these differences are significant or not.

Cow's milk: 1.90 1.95 2.00 2.02 1.85 1.80

Buffalo milk:2.12 2.00 2.20 2.45 2.20 2.10

H_0= Difference is not significant

An example

- $\bar{x}_1 = 1.92 \qquad \bar{x}_2 = 2.17 \quad s_1^2 = 0.00616$

$$s_2^2 = 0.0193617$$

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}\left[\frac{1}{n_1} + \frac{1}{n_2}\right]}}$$

If $|t|$<table value for 't' for a particular significance level and degree of freedom equal to $n_1 + n_2 - 2$?

Here $|t|$=3.6393 and degree of freedom=10. Let us take 95% confidence level. From table, the 't' value for degree of freedom 10 is 2.23, which is less than |t| obtained here. Hence, the difference is significant.

statistical process control (SPC)

- – Data figures
- – Pareto analysis
- – Cause-and-effect analysis
- – Trend analysis
- – Histograms
- – Scatter diagrams
- – Process control charts
- Identification and analysis of improvement opportunities

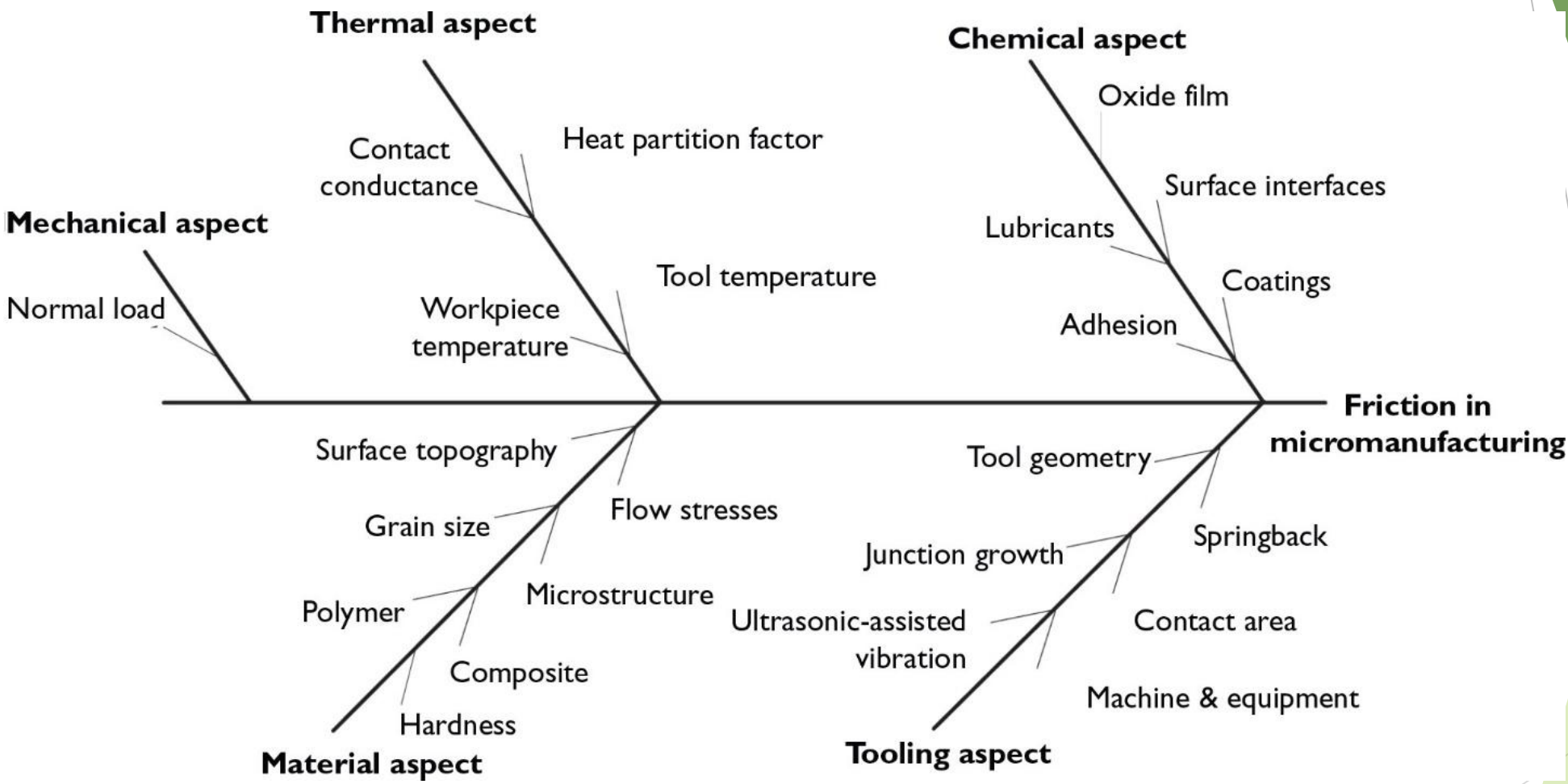
Data Tables

- Data tables (or data arrays) provide a systematic method for collecting and displaying data.
- Data tables sometimes take the form of manual check sheets.
- Simple to understand and straightforward.

Data table for suppliers A, B & C

Defect	A	B	C	Total
Incorrect Invoice	////	/		5
Incorrect Inventory	/////	//	/	8
Damaged material	///		//	5
Incorrect test documentation	/	///	////	8
Total	13	6	7	26

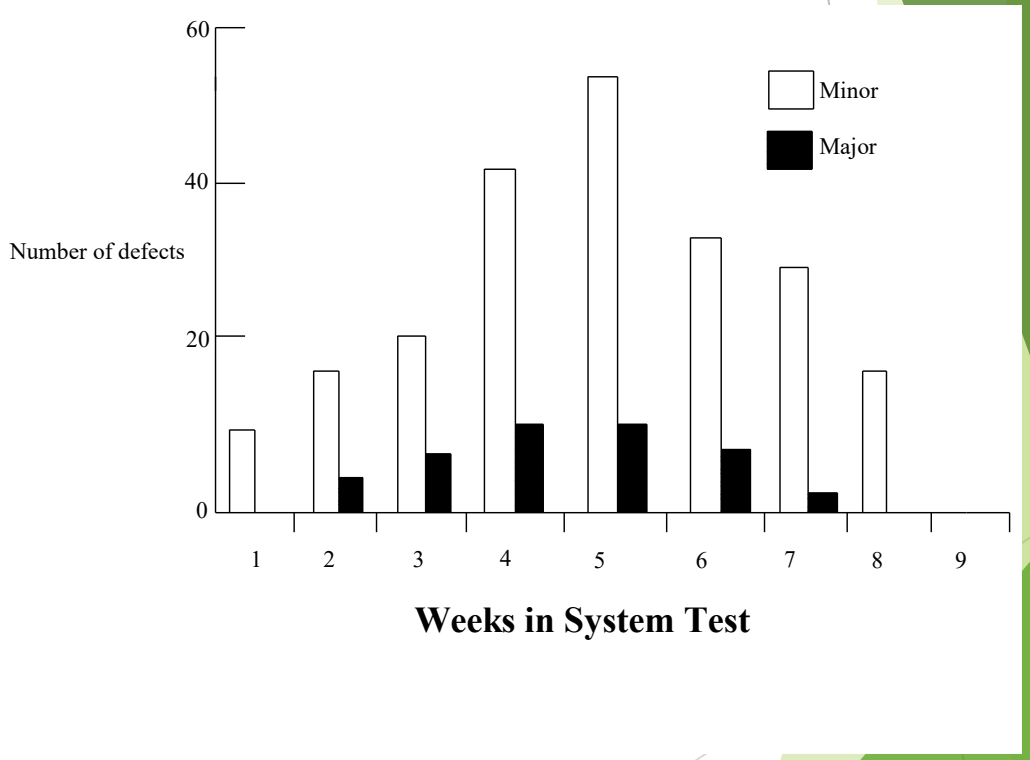
Cause and effect analysis



Dixit US, Yadav V, Ganesh Narayanan R, Bhardwaj N. Friction in micromanufacturing. *Journal of Micromanufacturing*. 2018;1(1):76-91

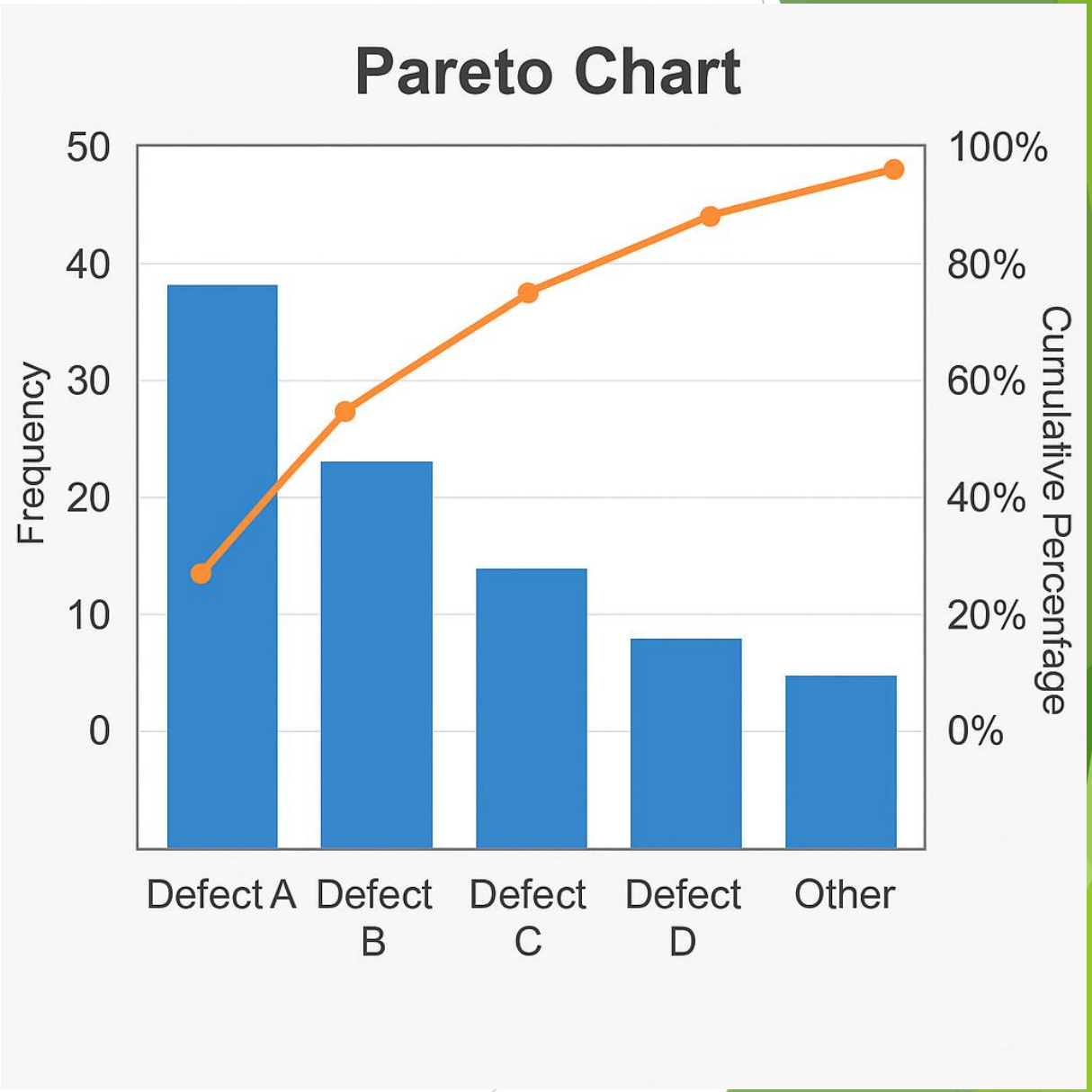
Histograms

- Graphical representation of data as a frequency distribution
- Quick look at the cumulative data at a single point in time
- No presentation of variance or trends over time



Pareto Analysis

- Specific type of histogram to identify and prioritize problem areas.
- The Pareto diagram directs attention to the most frequently occurring element.
- Three usages:
- Basic Pareto analysis identifies the vital few contributors that account for most quality problems in any system



Scatter Diagrams

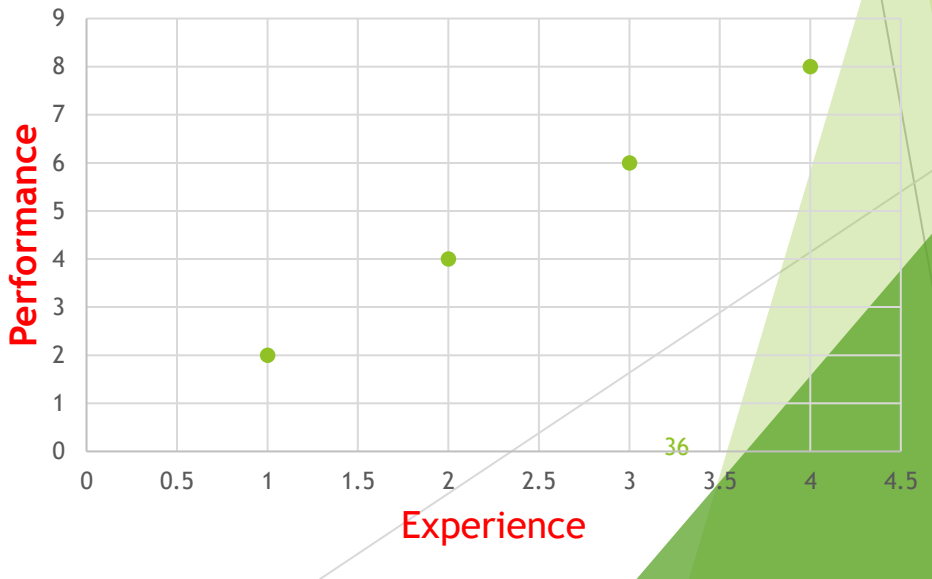
- Pictorial representation of process control data
- Organizes data using two variables: an independent variable and dependent variable

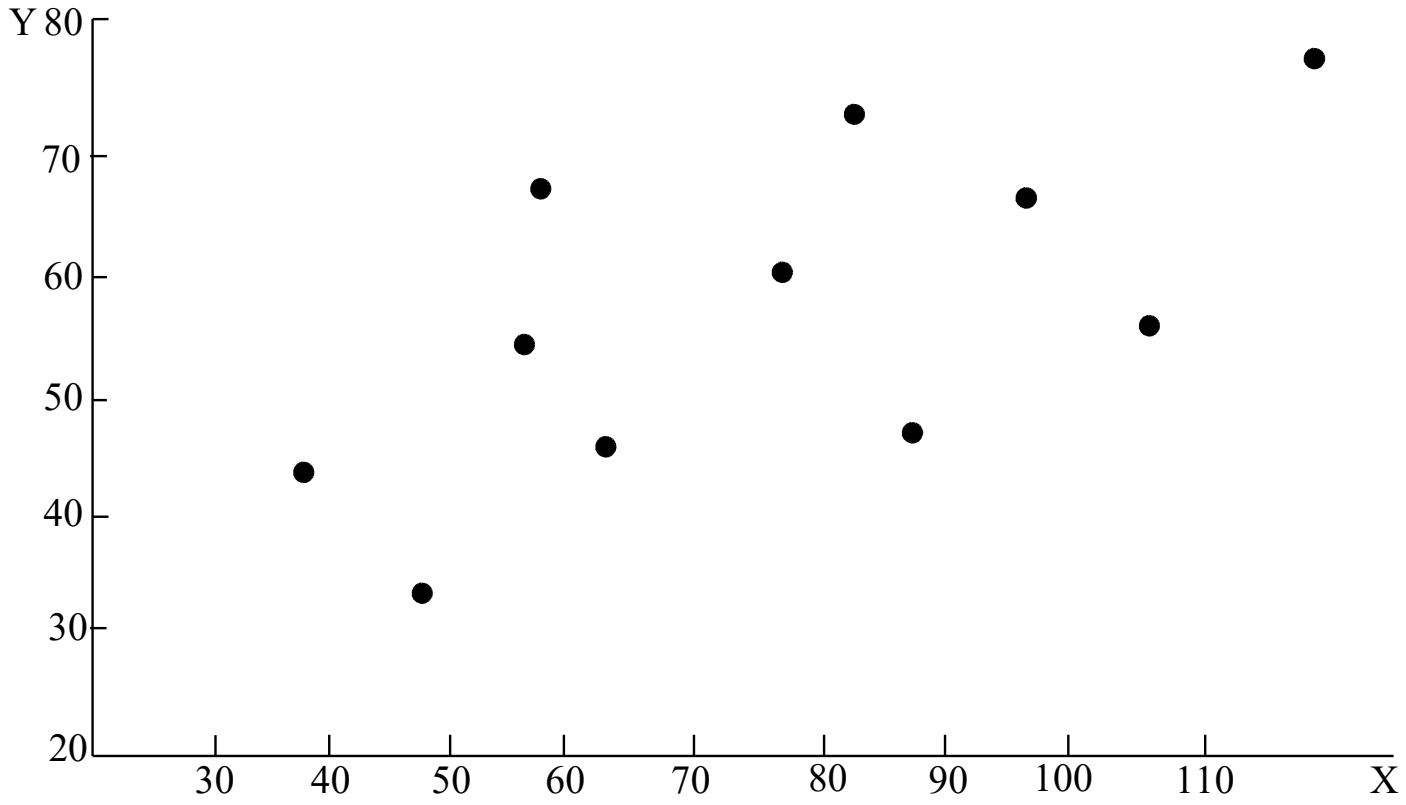
Example:

- – Independent variable: experience in months,
- – Dependent variable: performance

categories of relationships:

- – No correlation
- – Curvilinear correlation
- – Negative correlation
- – Positive correlation




Y
80
70
60
50
40
30
20
30
40
50
60
70
80
90
100
110
X

- Trend Analysis
- Statistical method for determining the equation that best fits the data in a scatter plot.
- Trend analysis provides forecasting to predict what is likely to occur in the future.

Control Charts

- Control charts focuses on the

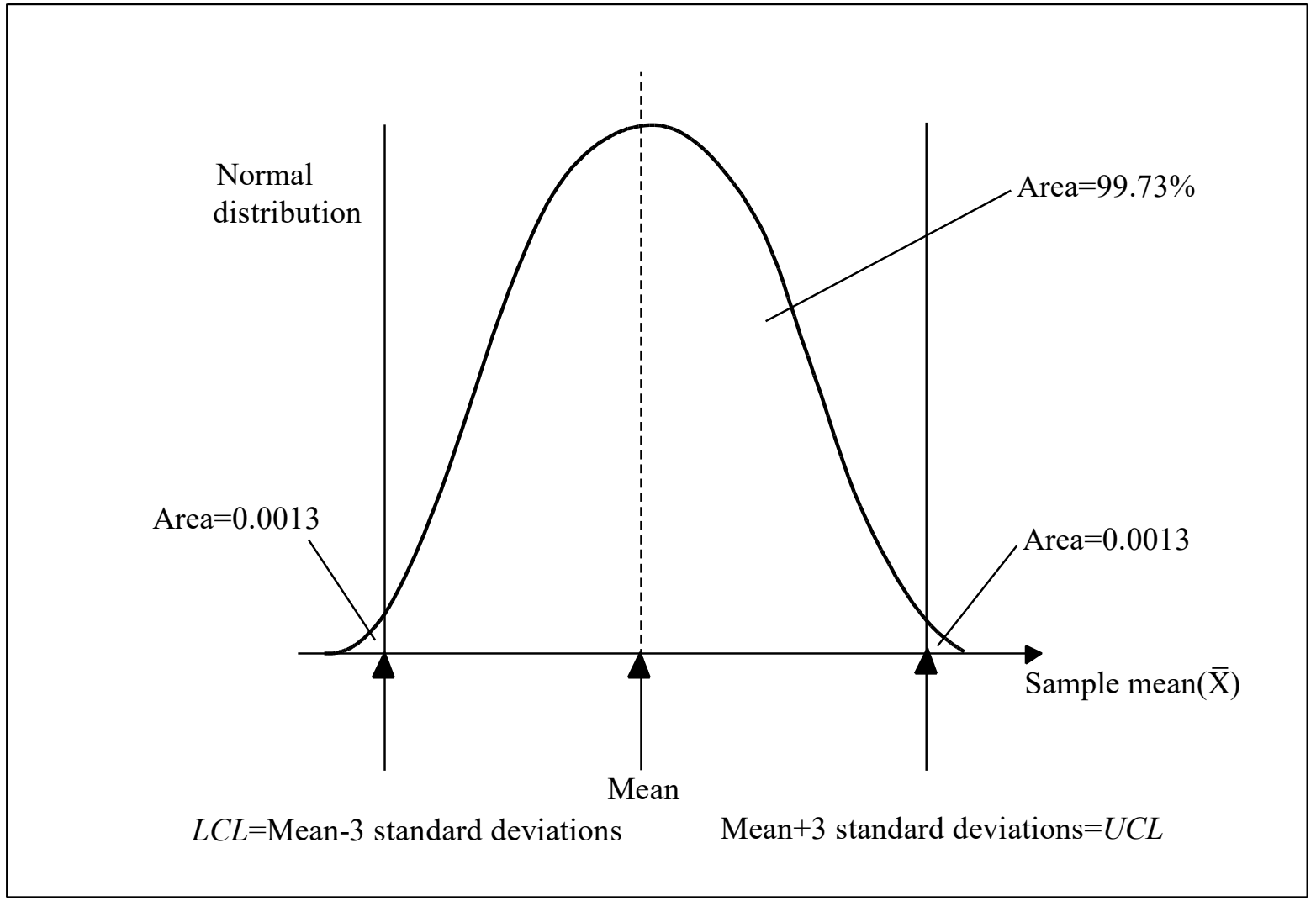
prevention of defects Based on the normal distribution

- Upper and lower control limits

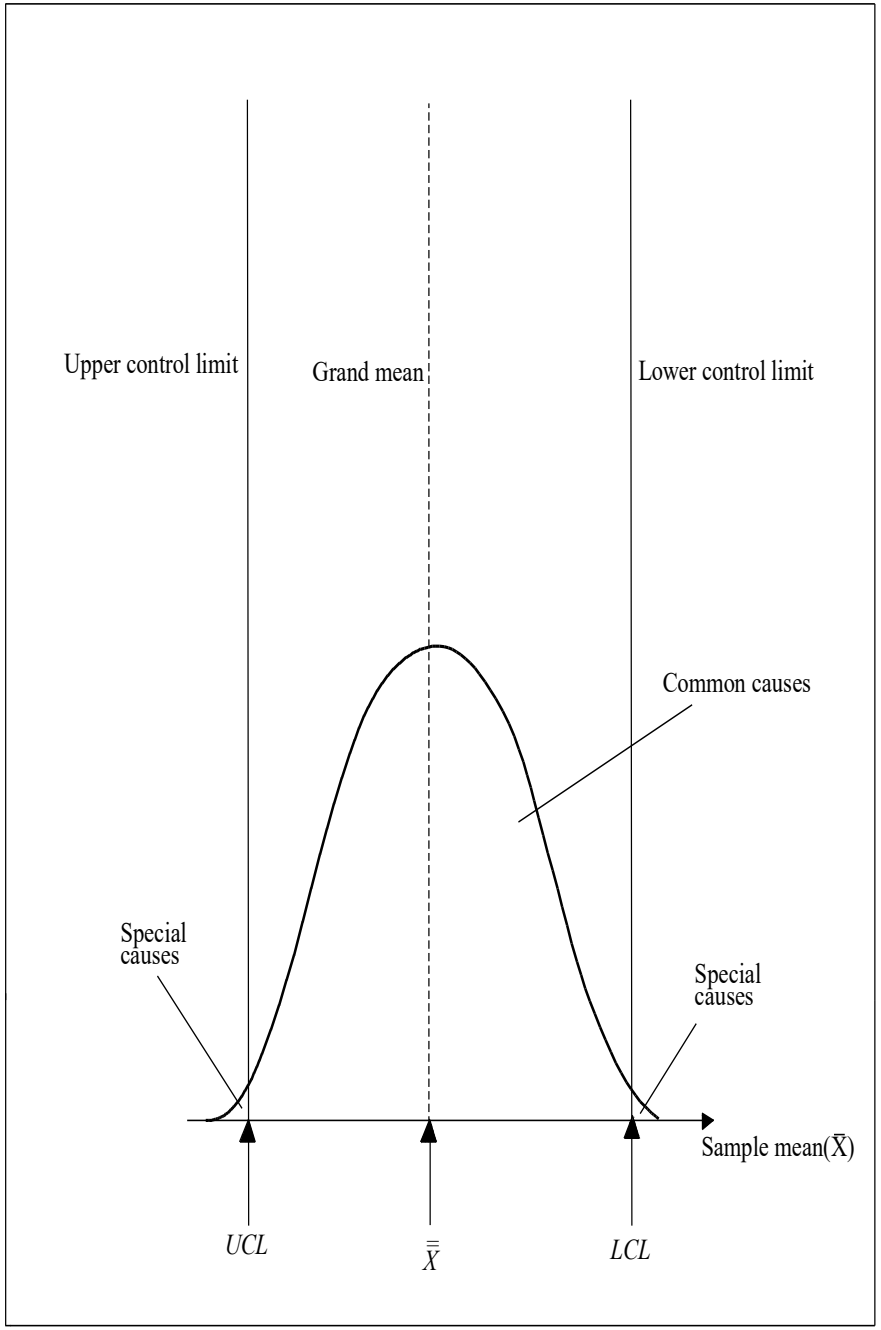
(UCL and LCL) represents this mean plus and minus three standard deviation of the data. 99.73% of all measurement lie with three-sigma limits.

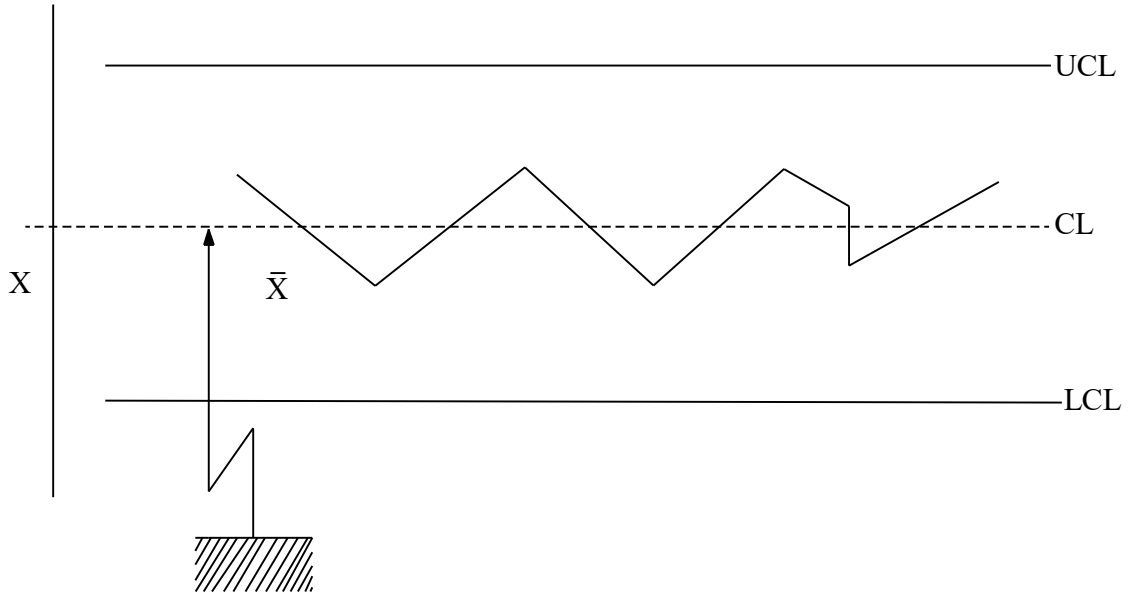
- Control chart analysis determines whether the inherent process variability and the process average are at stable levels, or whether appropriate action needs to be taken.

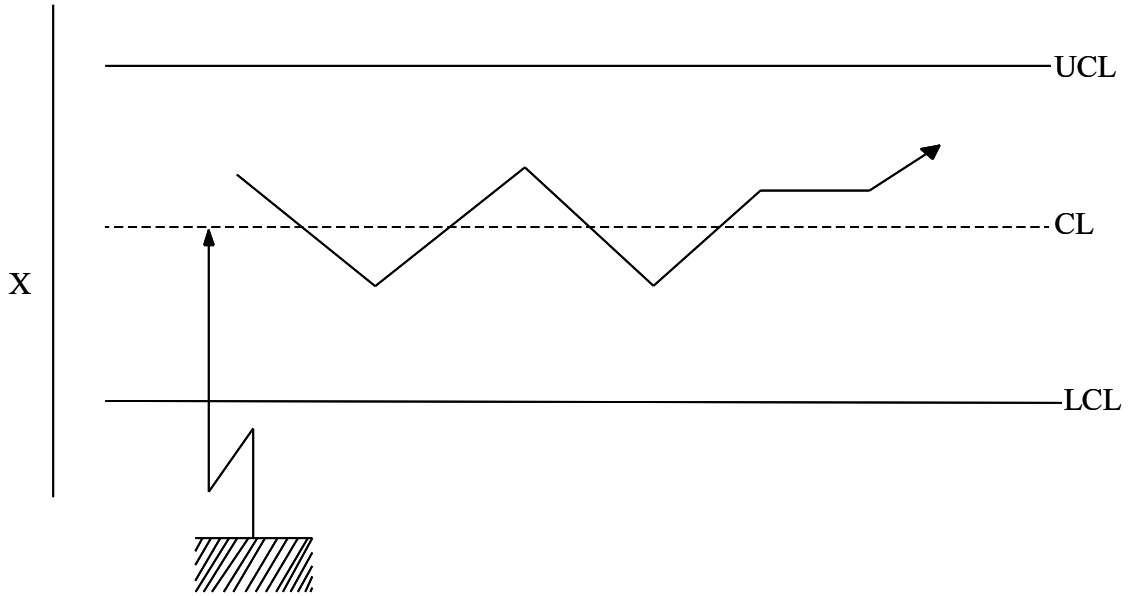
From this


Normal distribution
Area=99.73%
Area=0.0013
Area=0.0013
Sample mean($\bar{X}$)
Mean
LCL=Mean-3 standard deviations
Mean+3 standard deviations=*UCL*

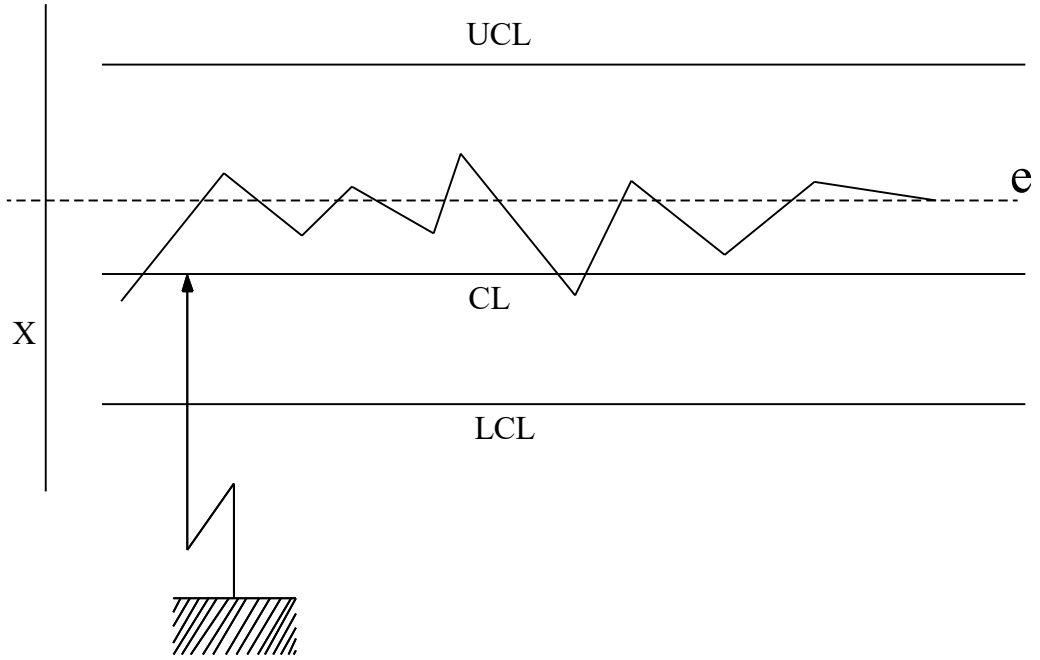
to this


Upper control limit
Grand mean
Lower control limit
Common causes
Special causes
Special causes
Sample mean($\bar{X}$)
UCL
$\bar{\bar{X}}$
LCL

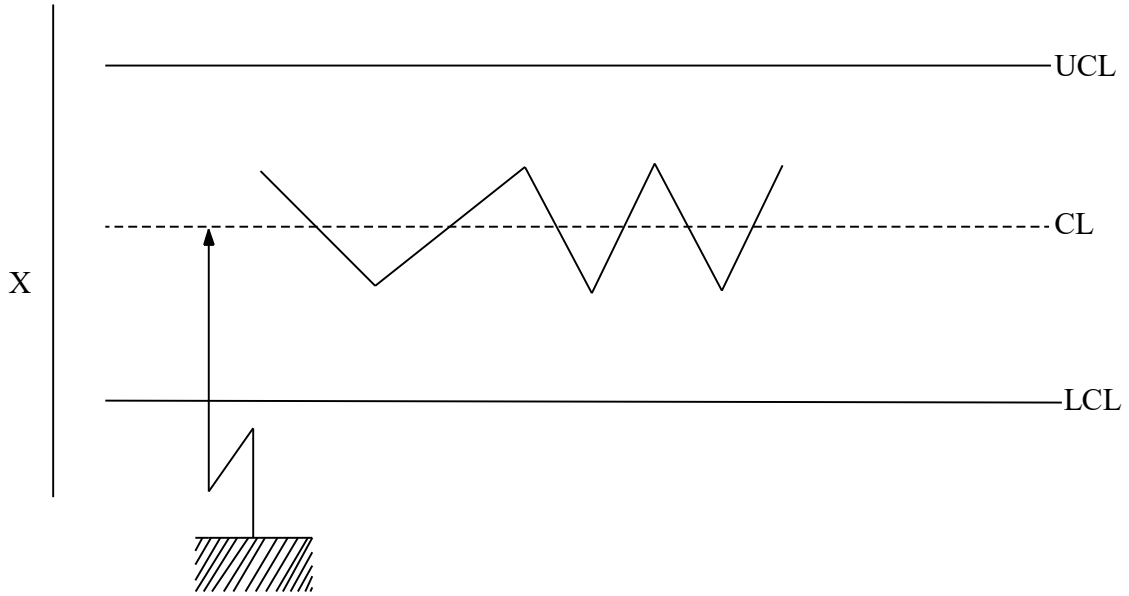

UCL
CL
LCL
X
$\bar{X}$



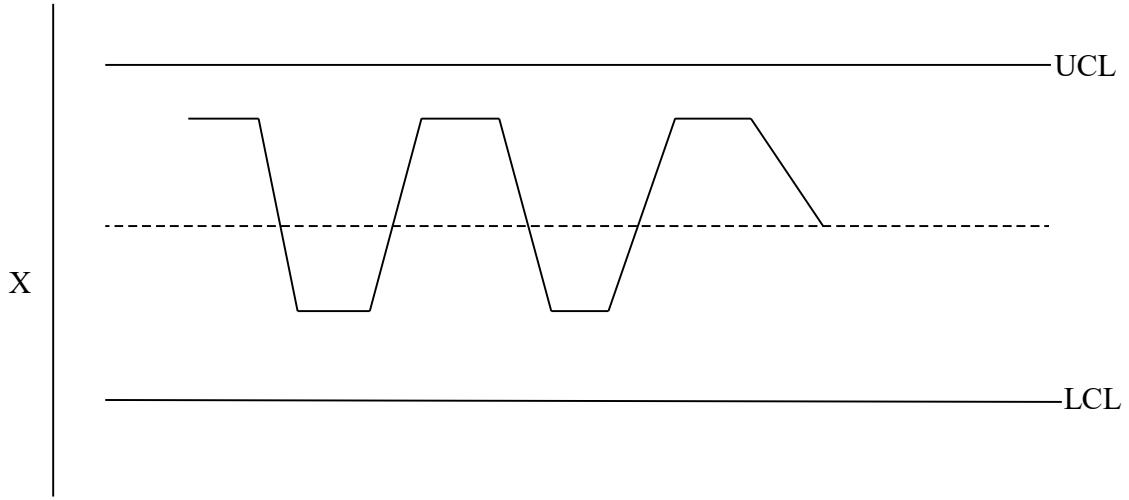
(a) Tool wear (Trend)



(b) Tool setting error (e) shift.



(a) Extreme variation.



(b) Cyclic variation.

- Process capability (*Cp*) is the ability to produce a product that conforms to design specification.
- $Cp = (USL – LSL) / 6\sigma$
- *USL: upper specification limit;*
- *LSL: lower specification limit*
- If $Cp > 1.33$, the process is well within the customer's specifications requirements